

One local factor: a proof of Mathar's Möbius-transform conjecture on OEIS A062368

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Abstract

OEIS A062368 is the multiplicative function with $a(p^e) = (e+1)(e+2)(4e+3)/6$. In August 2012 R. J. Mathar conjectured that it is the third inverse Möbius transform of $4^{\omega(n)}$, where ω counts distinct prime factors. We prove it. Both sides are multiplicative, so the whole statement is a single identity between local Dirichlet factors, and that identity is $\binom{e+3}{3} + 3\binom{e+2}{3} = (e+1)(e+2)(4e+3)/6$: the generating function of $a(p^e)$ is $(1+3x)/(1-x)^4$, which is exactly the local factor of 4^ω times three copies of ζ .

1 The sequence and the conjecture

OEIS A062368 (Vladeta Jovovic, 7 July 2001) has offset 1 and is defined as the multiplicative function with

$$a(p^e) = \frac{(e+1)(e+2)(4e+3)}{6}, \quad (1)$$

so $a = 1, 7, 7, 22, 7, 49, 7, 50, 22, 49, \dots$. The entry carries

*Conjecture: this is the third inverse Möbius transform of the sequence $4^{A001221(n)}$.
— R. J. Mathar, Aug 09 2012*

Here $A001221(n) = \omega(n)$ is the number of distinct primes dividing n , and the *inverse Möbius transform* of an arithmetic function b is $(b * \mathbf{1})(n) = \sum_{d|n} b(d)$, where $*$ is Dirichlet convolution and $\mathbf{1}$ is the constant function 1. Applying it three times gives $b * \mathbf{1} * \mathbf{1} * \mathbf{1} = b * \tau_3$, with $\tau_3 = \mathbf{1} * \mathbf{1} * \mathbf{1} = A007425$. So the conjecture reads

$$a = 4^\omega * \mathbf{1} * \mathbf{1} * \mathbf{1}, \quad \text{i.e.} \quad a(n) = \sum_{d|n} 4^{\omega(d)} \tau_3(n/d). \quad (2)$$

As of the “Last modified” line on the live entry (24 August 2026) the statement is still recorded as an unproven conjecture, with no proof in the entry’s links or programs.

Remark (status, stated plainly). On 10 July 2026 R. Oudra added to the FORMULA section of the same entry the line $a(n) = \sum_{d|n} 4^{\omega(d)} \tau_3(n/d)$, unlabelled and without proof, together with the Dirichlet generating function $\zeta(s)^4 \prod_p (1 + 3p^{-s})$. That formula is (2), i.e. Mathar’s conjecture restated; it was added six weeks before this note and the conjecture label was not removed. I have found no proof of either on the entry or in the literature. What follows supplies one, and shows the two statements are the same.

2 Proof

Write $x = p^{-s}$ and, for an arithmetic function b , let $B_p(x) = \sum_{e \geq 0} b(p^e)x^e$ denote its local factor at p ; for multiplicative b the Dirichlet series is $\sum_n b(n)n^{-s} = \prod_p B_p(p^{-s})$, and Dirichlet convolution corresponds to multiplication of local factors. All series below are formal.

Lemma 1. *The local factor of 4^ω at every prime is $(1 + 3x)/(1 - x)$, and hence the local factor of $4^\omega * \mathbf{1} * \mathbf{1} * \mathbf{1}$ is $(1 + 3x)/(1 - x)^4$.*

Proof. $\omega(p^e) = 1$ for $e \geq 1$ and $\omega(1) = 0$, so

$$\sum_{e \geq 0} 4^{\omega(p^e)} x^e = 1 + 4 \sum_{e \geq 1} x^e = 1 + \frac{4x}{1 - x} = \frac{1 + 3x}{1 - x}.$$

The local factor of $\mathbf{1}$ is $\sum_{e \geq 0} x^e = 1/(1 - x)$; multiplying by it three times gives the second claim. $\square$

Lemma 2. *The local factor of a at every prime is $(1 + 3x)/(1 - x)^4$.*

Proof. Since $(1 - x)^{-4} = \sum_{e \geq 0} \binom{e+3}{3} x^e$,

$$\frac{1 + 3x}{(1 - x)^4} = \sum_{e \geq 0} \left[\binom{e+3}{3} + 3 \binom{e+2}{3} \right] x^e,$$

with the convention $\binom{2}{3} = 0$ at $e = 0$. Now

$$\binom{e+3}{3} + 3 \binom{e+2}{3} = \frac{(e+1)(e+2)(e+3)}{6} + 3 \cdot \frac{e(e+1)(e+2)}{6} = \frac{(e+1)(e+2)[(e+3) + 3e]}{6} = \frac{(e+1)(e+2)(4e+3)}{6}$$

which is $a(p^e)$ by (1). $\square$

Theorem 3. $a = 4^\omega * \mathbf{1} * \mathbf{1} * \mathbf{1}$; that is, Mathar's conjecture holds:

$$a(n) = \sum_{d|n} 4^{\omega(d)} \tau_3(n/d) \quad \text{for every } n \geq 1.$$

Proof. Both sides are multiplicative — a by definition, and the right-hand side because Dirichlet convolution of multiplicative functions is multiplicative. Two multiplicative functions are equal precisely when they agree at every prime power, equivalently when their local factors agree at every prime. By Lemmas 1 and 2 both local factors equal $(1 + 3x)/(1 - x)^4$ at every prime. $\square$

Corollary 4. *The Dirichlet generating function of A062368 is $\zeta(s)^4 \prod_p (1 + 3p^{-s})$.*

Proof. Take the product of the local factors of Lemma 2 over all primes and use $\prod_p (1 - p^{-s})^{-1} = \zeta(s)$. $\square$

Remark (why the number three appears twice, and independently). The conjecture involves three copies of $\mathbf{1}$; the constant 3 in the local factor $(1 + 3x)$ comes from $4 - 1$, not from the number of transforms. They are unrelated, which is presumably why the identity looked opaque enough to be left as a conjecture: the statement pairs the multiplier 4 in 4^ω with the exponent 4 in $(1 - x)^{-4}$, and those two fours have different origins — one is $1 + 3$, the other is $1 + 3$ copies of ζ .

Remark (honest assessment). Half a page, and nothing beyond multiplicativity and one binomial identity. The natural destination is a comment on the entry. It is worth recording because the statement has carried the label “Conjecture” since 2012, and because the same identity was added to the same entry’s FORMULA section as an unlabelled assertion in July 2026 without the conjecture being marked resolved — the same pattern seen on A005329 and A129364, where a later contributor restates a standing conjecture as a fact and nobody reconciles the two.

3 Verification

A verification script, written separately from this note, checks every claim and prints every number quoted. It (i) reproduces the first 68 published terms of A062368 from the multiplicative definition (1); (ii) confirms the conjecture by applying the inverse Möbius transform to 4^ω three times, literally, and comparing with $a(n)$ for $1 \leq n \leq 300$, with no discrepancy, and likewise in the convolved form $\sum_{d|n} 4^{\omega(d)} \tau_3(n/d)$; (iii) confirms the local-factor identity of Lemma 2 symbolically, that the coefficients of $(1+3x)/(1-x)^4$ agree with $(e+1)(e+2)(4e+3)/6$ through x^{24} , and confirms $\binom{e+3}{3} + 3\binom{e+2}{3} = (e+1)(e+2)(4e+3)/6$ for $0 \leq e \leq 40$; and (iv) confirms that the threefold transform of 4^ω is multiplicative on all coprime pairs $m, n < 40$.

References

- [1] N. J. A. Sloane et al., *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org/A062368>, consulted 24 August 2026; also A001221 and A007425.
- [2] T. M. Apostol, *Introduction to Analytic Number Theory*, Springer, 1976; Chapter 2 (Dirichlet multiplication) and Chapter 11 (Dirichlet series and Euler products).

A Suggested OEIS comment (plain text, house style)

Mathar’s conjecture is true, and reduces to a single local identity. Both sides are multiplicative – $a(n)$ by definition, the transform because Dirichlet convolution preserves multiplicativity – so it is enough to compare local factors, i.e. to compare $\text{Sum}_{e \geq 0} b(p^e) x^e$ at each prime.

For 4^ω : $\omega(p^e) = 1$ for $e \geq 1$ and $\omega(1) = 0$, so the local factor is $1 + 4x/(1-x) = (1+3x)/(1-x)$. Convolving three times with the constant function 1 multiplies by $1/(1-x)$ three more times, giving $(1+3x)/(1-x)^4$.

For $a(n)$: since $(1-x)^{-4} = \text{Sum}_e C(e+3,3) x^e$,

$$(1+3x)/(1-x)^4 = \text{Sum}_e [C(e+3,3) + 3C(e+2,3)] x^e,$$

$$\begin{aligned} \text{and } C(e+3,3) + 3C(e+2,3) &= (e+1)(e+2)(e+3)/6 + 3e(e+1)(e+2)/6 \\ &= (e+1)(e+2)[(e+3) + 3e]/6 = (e+1)(e+2)(4e+3)/6 = a(p^e). \end{aligned}$$

The two local factors agree at every prime, so $a = 4^\omega * 1 * 1 * 1$, i.e.

$a(n) = \sum_{d|n} 4^{\omega(d)} \tau_3(n/d)$. This also gives the Dirichlet g.f.
 $\zeta(s)^4 * \prod_p (1 + 3/p^s)$ directly. Note the FORMULA line
 $"a(n) = \sum_{d|n} 4^{\omega(d)} \tau_3(n/d)"$ (R. Oudra, Jul 10 2026) is exactly
 this conjecture restated; the above proves both.